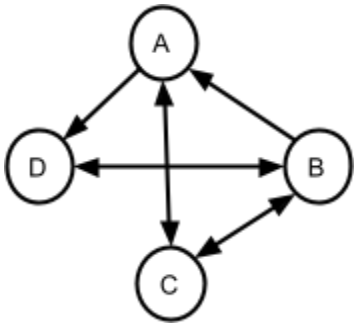
American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Senior Division

1. Graph Theory

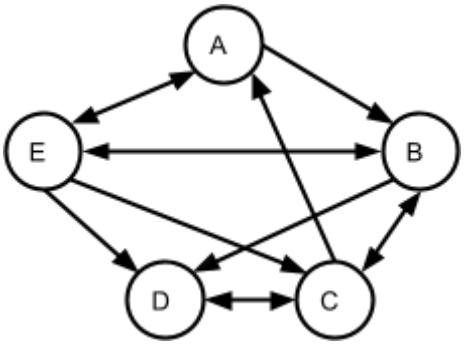
How many paths of length 4 are there from A in the directed graph below?



A. 11
B. 13
C. 16
D. 21
E. 64

2. Graph Theory

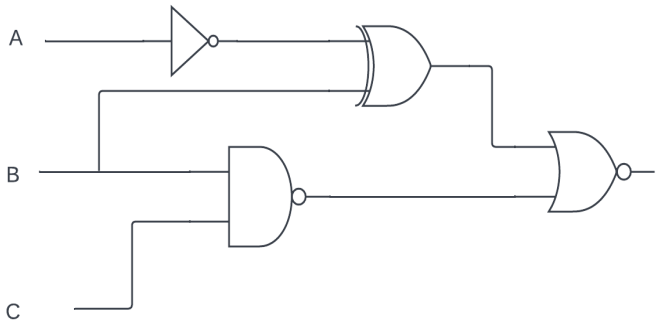
How many cycles from A are added to the directed graph below if the directed edge DE is added?



A. 1
B. 2
C. 3
D. 4
E. 5

3. Digital Electronics

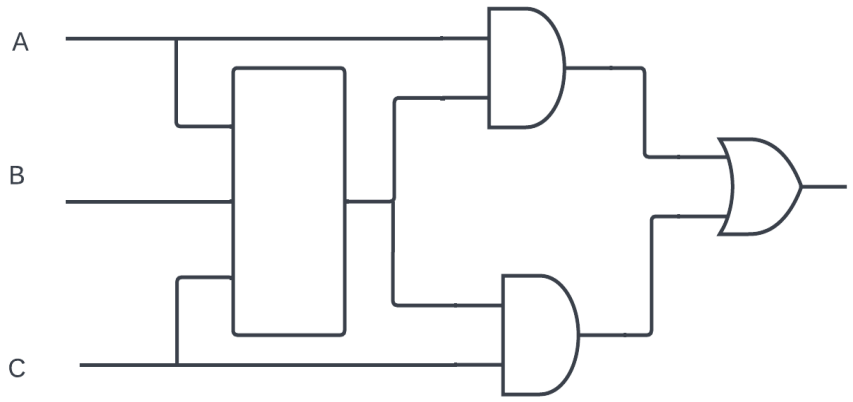
Simplify the Boolean expression represented by the following digital circuit:



A. ABC
B. $AB\overline{C}$
C. $A\overline{B}C$
D. $\overline{A}BC$
E. $\overline{A}B\overline{C}$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE if a new gate (rectangle) is defined to be TRUE if exactly 2 of the 3 inputs are TRUE?



A. 0
B. 1
C. 3
D. 5
E. 8

5. Assembly Language

What is output when the program is executed?

```
A       DC      0
B       DC      1
TOP     LOAD    B
        MULT    =2
        ADD     A
        STORE   C
        LOAD    =0
        STORE   D
TRY     LOAD    D
        ADD     =1
        STORE   D
        MULT    D
        SUB     C
        BE      STOP
        BL      TRY
        LOAD    B
        STORE   A
        LOAD    C
        STORE   B
        BU      TOP
STOP    PRINT   C
        END
```

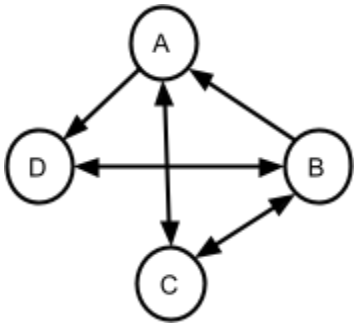
A. 29
B. 70
C. 121
D. 169
E. 408

American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Senior Division

1. 图论

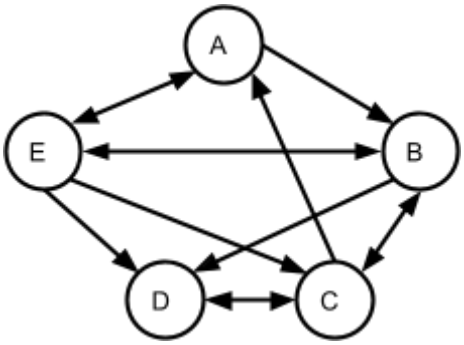
在下述有向图中，请问有多少条从 A 出发且长度为 4 的路径？



A. 11
B. 13
C. 16
D. 21
E. 64

2. 图论

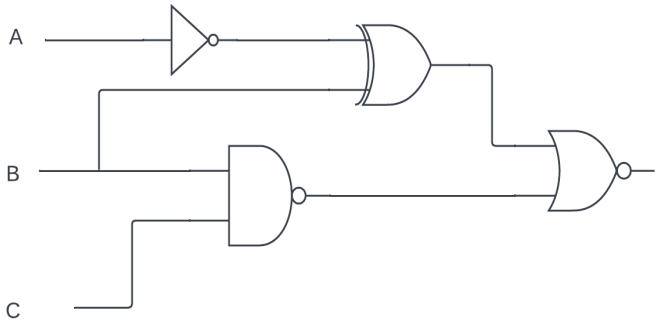
如果添加了有向边 DE，那么**下述**有向图**中**会增加多少个**从** A **出发的**环？



A. 1
B. 2
C. 3
D. 4
E. 5

3. 数字电路

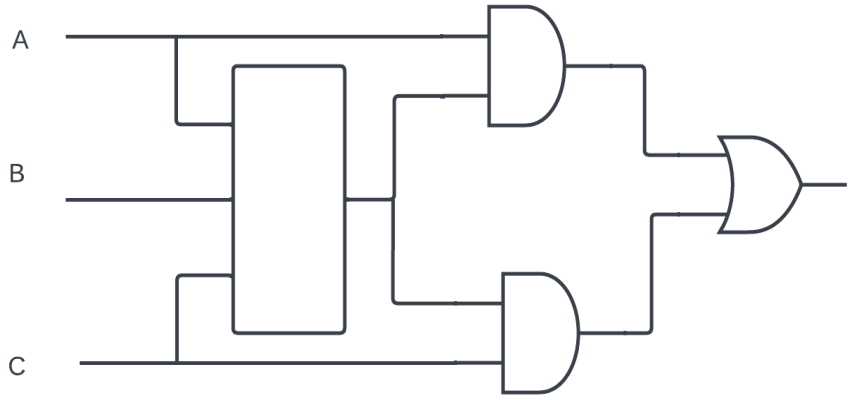
化简下述数字电路表示的布尔表达式：



A. ABC
B. $AB\overline{C}$
C. $A\overline{B}C$
D. $\overline{A}BC$
E. $\overline{A}B\overline{C}$

4. 数字电路

如果一个新的门（矩形）的 3 个输入中恰好有 2 个为**真**，则这个门被定义为**真**，那么有多少个有序三元组可以使以下数字电路为**真**？



A. 0
B. 1
C. 3
D. 5
E. 8

5. 汇编语言

执行以下程序**会**输出**什么**？

```
A       DC      0
B       DC      1
TOP     LOAD    B
        MULT    =2
        ADD     A
        STORE   C
        LOAD    =0
        STORE   D
TRY     LOAD    D
        ADD     =1
        STORE   D
        MULT    D
        SUB     C
        BE      STOP
        BL      TRY
        LOAD    B
        STORE   A
        LOAD    C
        STORE   B
        BU      TOP
STOP    PRINT   C
        END
```

A. 29
B. 70
C. 121
D. 169
E. 408